Modern Academy
for Engineering and Technology in Maadi
Electronic Engineering and
Communications Technology Department

Academic Year	2020 – 2021
Semester	Fall
Exam Date	4/3/2021

QUESTIONS FOR THE FINAL WRITTEN EXAM

Subject: (ELC 422) Digital Signal Processing
Examiners: Dr. Samir M Kamal & Dr. Sara M Hassan
Spec.: Level 4 Elec. Eng. & Comm. Tech. Dep.
Time: 2 hours

Number of Pages: 1 Number of Questions: 4 Attempt all questions

Question 1 **(15 points)**

a- In A/D, in detail, explain the quantization process of continuous-amplitude signals.
b- Consider the analog signal: $x_a(t) = 5\ (\cos 3000\pi t)$.
(i) Check the periodicity of this signal. If it is periodic, find its fundamental period.
(ii) Assume that, this signal is sampled at the rate $F_s = 9000$ Hz. What is the discrete-time signal obtained after sampling, x(n)? , Calculate the samples values of the first cycle in x(n).

Question 2 **(15 points)**

a- Define the convolution operation and convolution theorem.
b- For the following signal, x(n) = {1, 0, 1, 1}:
(i) Using the FFT method find the frequency components, X(k), and check your results by calculating x(n) using the IDFT algorithm.
(ii) Using your results, verify Parseval's theorem.
(iii) Plot the amplitude and the phase spectrums.

Question 3 **(10 points)**

a- Define the Auto-correlation operation and cross-correlation coefficient.
b- For the two sequences, x1 (n) & x2 (n) given by, [x1 (n) = {3, 2, 0, 1} & x2 (n) = {1, 0, 2, 4}] calculate:
(i) Their cross-correlation, then plot the obtained correlation function against lag number, j
(ii) The auto-correlation of the first sequence.

Question 4 **(20 points)**

a- For LTIS define the following items: Causality and difference equation.
b- Using the partial fraction method, find the impulse response, h(n), of the system characterized by the following transfer function, H(z).
Note: Use the table given at the exam end.

$$H(z) = \frac{(z+1)}{(z-0.5)(z-1)^2}$$

(i) Plot its pole – zero diagram,
(ii) Find its region of convergence,
(iii) Discuss the stability of this system,
(iv) Find the difference equation of this system, y (n) and
(v) Draw its realization diagram using signal flow diagram.

Z-transforms of some required sequences.

Discrete-time sequence x(n), $n \geq 0$	z-transform X(z)	Note
kn	$\frac{kz}{(z-1)^2}$	k and α are constants
k	$\frac{kz}{z-1}$	
$k\alpha^n$	$\frac{kz}{z-\alpha}$	

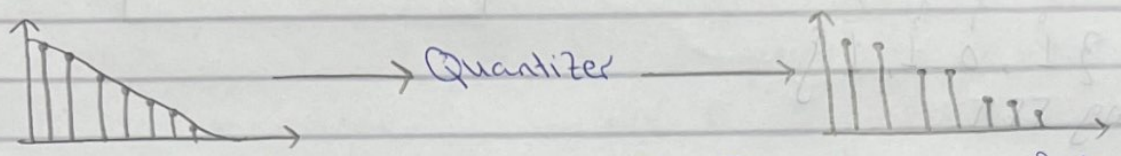
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Q1 :- a) Quantization process

- Convert Continuous amplitude into discrete amplitude by representing each sample to the nearest level.



[Discrete time & Continuous amplitude] [Discrete time & Amplitude]

Quantization error :- The difference between the quantized sample & the quantized o/p. $e_q(n) = X_a(n) - X(n)$ $e_q(n)$

b) $X_a(t) = 5\cos(3000\pi t)$

i) $X(t) = A\cos(2\pi ft + \theta)$ $\quad 2\pi f = 3000\pi \rightarrow f = \frac{3000}{2} = 1500$

$F = \frac{3000}{2} = 1500\ (Hz) \rightarrow t_p = \frac{1}{1500} = 6.6 * 10^{-4}$ sec

[Fundamental period]

Replace each $t \rightarrow t + t_p$

$x(t+t_p) = 5\cos(3000\pi(t+t_p))$

$= 5\cos(3000\pi t + 3000\pi t_p)$

$= 5\cos(3000\pi t + 3000\pi/1500)$

$= 5\cos(3000\pi t + 2\pi)$

$x(t+t_p) = 5\cos(3000\pi t) = x_a(t)$

Signal is periodic every $t_p = 1/500$ (sec)

Replace each $t \rightarrow nts$ $\quad 1/f_s$

$x(nts) = 5\cos(3000\pi n ts)$

$x(nts) = 5\cos(3000\pi \frac{n}{9000})$

$= \boxed{5\cos(\frac{\pi n}{3})}$

ii) $f_s = 9000$ Hz $\quad f_s \geq 2F \rightarrow 9000 \geq 2*1500 \rightarrow 9000 > 3000$ ✓

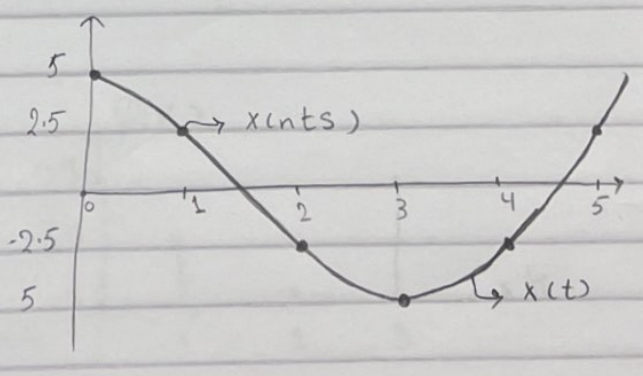
$F = \frac{f}{f_s} = \frac{1500}{9000} = \frac{K}{N_p} = \frac{1}{6}$

$N_p = 6$ samples

$X(0) = 5 \quad X(3) = -5$

$X(1) = 2.5 \quad X(4) = -2.5$

$X(2) = -2.5 \quad X(5) = 2.5$



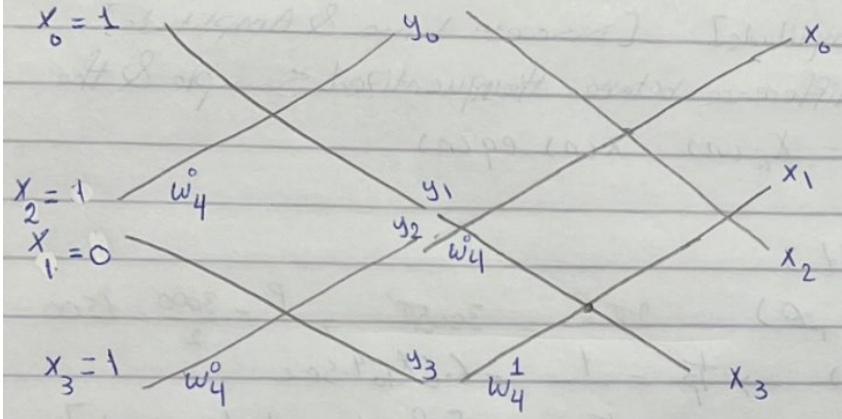
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Q2:- a) Convolution operation:- it Describes how the i/p interacts with the system to produce the o/p.

$x(n) \rightarrow$ [Impulse response $h(t)$] $\rightarrow y(n)$

b) $x(n) = \{\overset{x_0}{\underset{}{1}}, \overset{x_1}{0}, \overset{x_2}{1}, \overset{x_3}{1}\}$

i) By using FFT



$w_4^{-1} = -j$

$y_0 = x_0 + x_2 = 2$
$y_1 = x_0 - x_2 = 0$
$y_2 = x_1 + x_3 = 1$
$y_3 = x_1 - x_3 = -1$

$x(0) = y_0 + y_2 = 3$
$x(1) = y_1 + (-j)y_3 = +j$
$x(2) = y_0 - y_2 = 1$
$x(3) = y_1 - (-j)y_3 = -j$

$x(k) = \{3, j, 1, -j\}$

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By using IDFT $X(K) = \{3, j, 1, -j\}$ $N = 4$, $K = 0, 1, 2, 3$

$$x(n) = \frac{1}{N} \sum_{K=0}^{N-1} X(K)\, e^{j2\pi \frac{nK}{N}} = \frac{1}{4} \sum_{K=0}^{3} X(K)\, e^{j\pi \frac{nK}{2}}$$

$$x(n) = \frac{1}{4}\left[X(0) + X(1)\, e^{j\pi \frac{n}{2}} + X(2)\, e^{j\pi n} + X(3)\, e^{j\pi \frac{3n}{2}} \right]$$

$$= \frac{1}{4}\left[3 + (j)\, e^{j\pi \frac{n}{2}} + e^{j\pi n} + (-j)\, e^{j\pi \frac{3n}{2}} \right]$$

At $n(0) = 1$

At $n(1) = 0$

At $n(2) = 1$

At $n(3) = 1$

$$x(n) = \{1, 0, 1, 1\}$$

ii) Parseval's theorem :-

$$\sum_{n=0}^{N-1} x^2(n) = \frac{1}{N} \sum_{K=0}^{N-1} |X(K)|^2$$

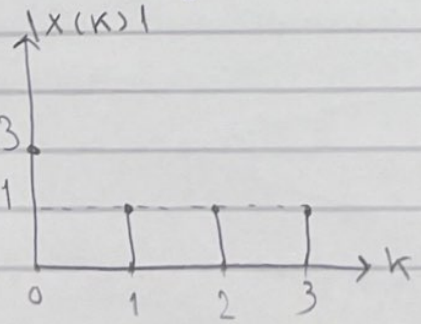
$$|X(K)| = \sqrt{Re^2 + img^2} = \{3, 1, 1, 1\}$$

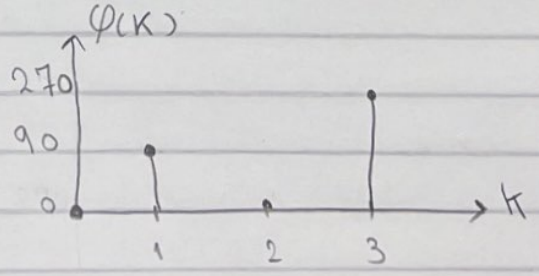
L.H.S $= 3$, R.H.S $= \frac{1}{4} * 3^2 + 1^2 + 1^2 + 1^2 = 3$

Parseval's theorem is verified.

iii)

$$\varphi(K) = \{0°, 90°, 0°, 270°\}$$





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Q3 :- a) Auto correlation operation :

It measure the degree of similarity of the same signal

$$r_{11}(j) = \frac{1}{N_{new}} \sum_{n=0}^{N_{new}-1} x_1(n)\, x_1(n+j)$$

Cross Correlation coefficient :

it measures cross correlation according to the fixed scale between -1 & 1

+1 → 100 % correlation 'identical signals'

-1 → 100 % correlation 'inverse signals'

0 → No correlation as signals are independent from each other

$$\rho_{12}(j) = \frac{r_{12}(j)}{\frac{1}{N_{new}} \sqrt{\sum_{n=0}^{N_{new}-1} x_1^2(n) \sum_{n=0}^{N_{new}-1} x_2^2(n)}}$$

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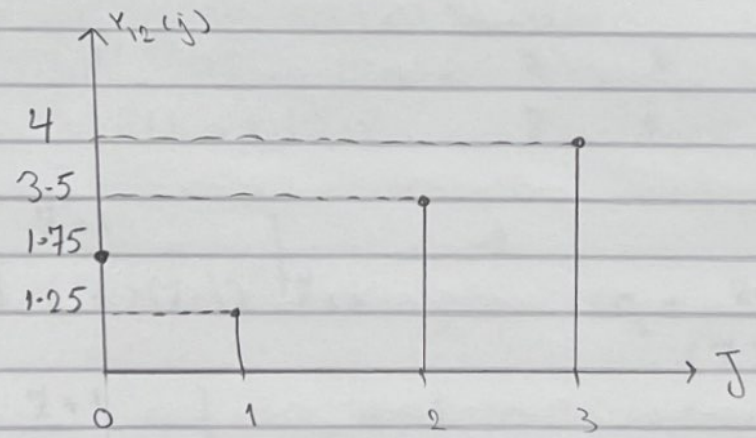
b) $x_1 = \{3, 2, 0, 1\}$ $x_2 = \{1, 0, 2, 4\}$

i) By using cross correlation :- 'cyclic'

$N_1 = 4$ $N_2 = 4$ $N_{new} = 4$

$x_1(n)$	3	2	0	1	J	$r_{12}(j)$	$\rho_{12}(j)$
	(1)	0	2	4	0	1.75	0.408
$x_2(n+j)$	(0)	2	4	1	1	1.25	0.291
	(2)	4	1	0	2	3.5	0.816
	4	1	0	2	3	4	0.933

$$\rho_{12}(j) = \frac{r_{12}(j)}{\frac{1}{N_{new}}\sqrt{\sum_{n=0}^{N_{new}-1} x_1^2(n) \sum_{n=0}^{N_{new}-1} x_2^2(n)}} = \frac{r_{12}(j)}{\frac{1}{4}\sqrt{14 \times 21}} = \frac{r_{12}(j)}{4.2866}$$



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ii) By using Auto - Correlation $x_1(n) = \{3, 2, 0, 1\}$ $N_{new} = 4$

$x_1(n)$					j	$r_{11}(j)$	$\rho_{11}(j)$
3	2	0	1		0	3.5	1
2	0	1	0		1	1.5	0.428
0	1	0	0		2	0.5	0.142
1	0	0	0		3	0.75	0.214

$$\rho_{11}(j) = \frac{r_{11}(j)}{\frac{1}{N_{new}} \sum_{0}^{N_{new}-1} x_1^2(n)} = \frac{r_{11}(j)}{\frac{1}{4} * 14} = \frac{r_{11}(j)}{3.5}$$

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Q4 :- a) Causality :- The i/p relationship for a LTIS is given by the convolution sum

$$y(n) = \sum_{k=-\infty}^{\infty} h(k) \cdot x(n-k)$$

Difference equation :- it represents the relation between the system response & an i/p in time domain

b) $$H(z) = \frac{z+1}{(z-0.5)(z-1)^2}$$

i) Poles : $z = \{0.5, 1, 1\}$
Zeros : $z = \{-1\}$

ii) Roc :- the entire z-plane except at $z = \{0.5, 1\}$

impulse response by partial fraction :-

$$\frac{H(z)}{z} = \frac{z+1}{z(z-0.5)(z-1)^2} = \frac{A}{z} + \frac{B}{z-0.5} + \frac{C_2}{(z-1)^2} + \frac{C_1}{(z-1)}$$

$$A = \left.\frac{z+1}{(z-0.5)(z-1)^2}\right|_{z=0} = -2$$

$$C_2 = \left.\frac{z+1}{(z)(z-0.5)}\right|_{z=1} = 4$$

$$B = \left.\frac{z+1}{z(z-1)^2}\right|_{z=0.5} = 12$$

$$C_1 = \frac{d}{dz}\left[\frac{z+1}{z(z-0.5)}\right] = \left.\frac{(z^2-0.5z)-(z+1)(2z-0.5)}{(z^2-0.5z)^2}\right|_{z=1} = -10$$

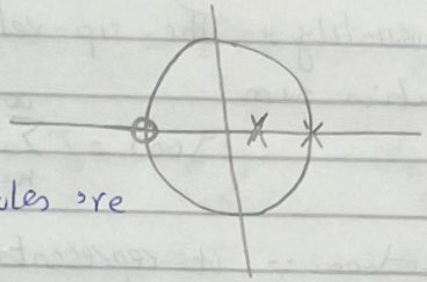
$$H(z) = -2 + \frac{12z}{z-0.5} + \frac{4z}{(z-1)^2} - \frac{10z}{z-1}$$

$$h(n) = -2\delta(n) + 12(0.5)^n + 4n - 10(1)^n$$

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o → Zeros
x → poles

iii) System stability.



System is marginally stable is there's one pole on the unit circle & no poles are outside

iv) Difference equation

$$H(z) = \frac{(z+1)}{(z-0.5)(z-1)^2} = \frac{(z+1)}{(z-0.5)(z^2-2z+1)}$$

$z^3 - 2z^2 + z - 0.5z^2 + z - 0.5$

$$= \frac{z+1}{z^3 - 2.5z^2 + 2z - 0.5} * \frac{z^{-3}}{z^{-3}}$$

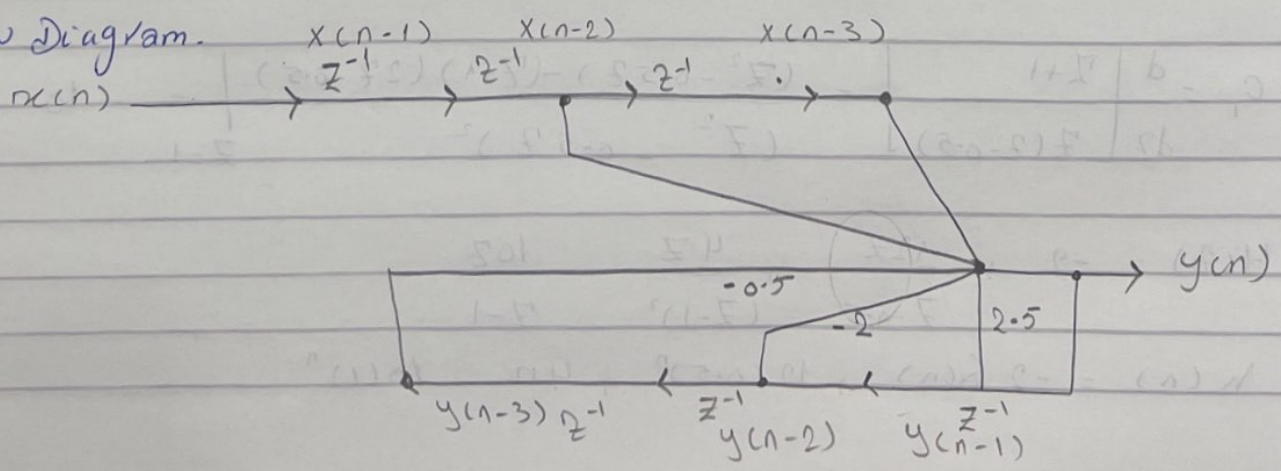
$$= \frac{z^{-2} + z^{-3}}{1 - 2.5z^{-1} + 2z^{-2} - 0.5z^{-3}} = \frac{Y(z)}{X(z)}$$

$$Y(z) - 2.5Y(z)z^{-1} + 2Y(z)z^{-2} - 0.5Y(z)z^{-3} = X(z)z^{-2} + X(z)z^{-3}$$

$$y(n) - 2.5y(n-1) + 2y(n-2) - 0.5y(n-3) = x(n-2) + x(n-3)$$

$$\rightarrow y(n) = x(n-2) + x(n-3) + 2.5y(n-1) - 2y(n-2) - 0.5y(n-3)$$

v) Flow Diagram.



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